

Week 1 — Phase 1

Understanding the Nigeria DHS Dataset

Before touching a single line of analysis, every researcher should answer five questions:

1. What is this dataset?
2. Who was measured?
3. How were the variables collected?
4. What assumptions must we respect?
5. What are the limitations?

This phase is often skipped by beginners, but it is what separates data analysts from data scientists.

1. What is the NDHS?

The **Nigeria Demographic and Health Survey (NDHS)** is one of the world's most widely used public health datasets. It is part of the global DHS Program, which collects nationally representative demographic and health information using standardized protocols.

For this assignment, we are using the **Children's Recode (KR) file**, where **each row corresponds to one child born within the five years preceding the survey**. The dataset links each child's information to the mother's interview, enabling analysis of child health alongside maternal, household, and environmental characteristics.

2. Why this dataset is important

Unlike toy datasets such as Iris or Titanic, NDHS is a **real-world epidemiological dataset**.

Researchers worldwide use DHS data to study:

- malnutrition
- vaccination coverage
- maternal health
- fertility
- mortality
- poverty
- education
- sanitation

- inequalities

Many WHO and UNICEF reports are based on DHS analyses.

Learning to analyse DHS data is therefore a valuable research skill.

3. Unit of Analysis

One of the most common mistakes beginners make is misunderstanding what one observation represents.

Here:

One row = one child

NOT

- one household
- one mother
- one hospital
- one community

This distinction is critical because several children from the same household may appear in the dataset.

4. Variables for Week 1

The assignment focuses on **15 variables**.

Variable	Description	Data Type	Role
hw71	Weight-for-age Z-score (WAZ)	Continuous	Outcome
hw72	Weight-for-height Z-score (WHZ)	Continuous	Outcome
hw1	Child age (months)	Continuous	Predictor
b4	Child sex	Categorical	Predictor
b5	Child survival status	Binary	Descriptive
m4	Breastfeeding status	Categorical	Predictor
v024	Region	Categorical	Predictor
v025	Urban/Rural	Binary	Predictor

v106	Mother's education	Ordinal	Predictor
v701	Father's education	Ordinal	Predictor
v130	Religion	Nominal	Predictor
v190	Wealth index	Ordinal	Predictor
v136	Household size	Continuous	Predictor
v137	Number of under-5 children	Continuous	Predictor
v005	Survey weight	Numeric	Sampling weight

These variables were selected because they capture biological, socioeconomic, demographic, and household factors associated with child nutrition.

5. The Three Outcome Variables

Although the assignment centres on **WAZ** and **WHZ**, understanding all three anthropometric indicators is essential.

Indicator	Measures	Threshold
HAZ	Height-for-age	Chronic malnutrition (stunting)
WAZ	Weight-for-age	Underweight
WHZ	Weight-for-height	Acute malnutrition (wasting)

A child may be:

- stunted only,
- wasted only,
- underweight only,
- or combinations of these conditions.

Each indicator reflects a different nutritional problem.

6. Understanding Z-scores

Many students find Z-scores intimidating, but the concept is straightforward.

A Z-score measures how far a child's measurement is from the healthy reference population.

Examples:

Z-score	Interpretation
0	Average child
+1	Above average
-1	Slightly below average
-2	Malnourished (clinical cutoff)
-3	Severe malnutrition

The DHS stores these values multiplied by **100**.

Therefore:

Stored value	Actual value
-165	-1.65
-243	-2.43
58	0.58

This scaling preserves precision while using integer storage. The first cleaning step is therefore to divide by **100**.

7. The DHS Missing-Value Trap

This is arguably the most important data quality issue in Week 1.

Unlike many datasets, DHS does **not** leave missing numeric values blank. Instead, it uses reserved codes.

For anthropometric variables:

Value	Meaning
9996	Missing
9997	Missing
9998	Missing
9999	Missing

If these are treated as genuine measurements, summary statistics, histograms, and correlations become severely distorted. Both the assignment and the diagnostic guide emphasise that these values must be recoded to NA before analysis.

8. Why hw73 Matters

A unique feature of DHS anthropometric data is the **biological plausibility flag** (hw73).

This variable identifies measurements considered biologically implausible by DHS quality-control procedures.

The recommended workflow is:

- Recode 9996–9999 to NA.
- Exclude records where $hw73 \neq 0$ for anthropometric analyses.

This is preferable to relying solely on generic statistical outlier rules because it incorporates domain expertise into data cleaning.

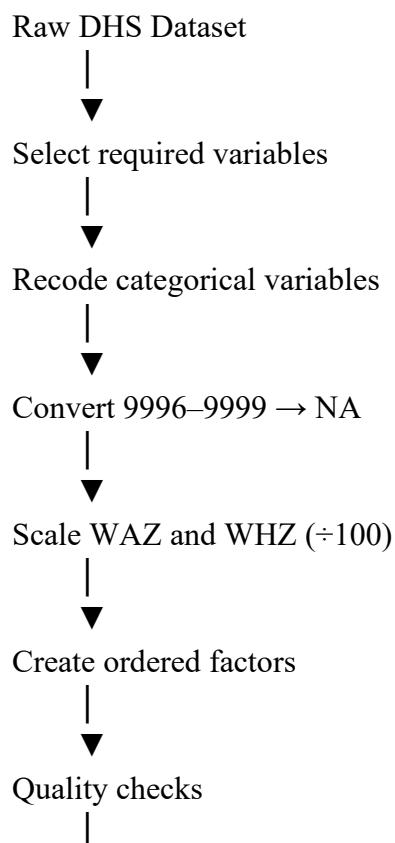
9. Survey Weights (v005)

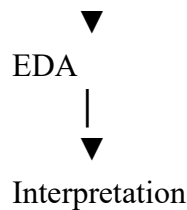
One variable that deserves special attention is **v005**, the survey sampling weight.

The assignment includes it among the selected variables, but the lecture examples do not use weighted analyses. For this sprint, we will retain v005 in the cleaned dataset without applying survey-weighted estimation, while noting this as a limitation. In later weeks, particularly when estimating national prevalence or fitting inferential models, incorporating survey weights will become important.

10. Expected Data Cleaning Pipeline

Before any statistics or plots, our pipeline will be:





This mirrors the standard EDA workflow advocated in the diagnostic guide: inspect the data, diagnose missingness, examine distributions, assess outliers, and then explore relationships.

Researcher's Note

Before we begin coding, one methodological point deserves attention.

The assignment asks us to use **15 variables**, but from a research perspective we will **not treat all variables equally**. Instead, we'll classify them as:

- **Outcomes:** WAZ and WHZ.
- **Primary explanatory variables:** wealth, education, region, residence, breastfeeding, religion.
- **Control/context variables:** age, sex, household size, number of under-5 children.
- **Survey design variable:** sampling weight (v005).

This conceptual separation will make our analyses, interpretations, and later research questions much stronger.

We will now proceed to **Phase 2: Data Preparation**, where we'll build a robust, reproducible R data-cleaning pipeline that fully satisfies **Task A** while following research and software engineering best practices rather than the minimum assignment requirements.

Phase 2 — Task A: Data Preparation

Objective

Build a **research-quality** data preparation pipeline that:

- satisfies the ADA assignment requirements,
- follows reproducible programming practices,
- can be reused in future projects.

Step 2.1 — Required R Packages

Rather than loading packages throughout the script, load everything once at the top.

```
#####
# ADA Global Academy
# Sprint 01 - Exploratory Data Analysis
# Nigeria DHS 2024
# Basil Emeokoro
#####
```

```
library(tidyverse)
library(readr)
library(skimr)
library(e1071)
library(corrplot)
library(knitr)
library(kableExtra)
```

Notice that **install.packages()** is intentionally omitted, consistent with the assignment instructions.

Step 2.2 — Read the Dataset

```
#####
# Read Dataset
#####

ndhs <- read_csv("NGKR8BFL.csv",
                 show_col_types = FALSE)
```

```
dim(ndhs)
```

```
glimpse(ndhs)
```

The `glimpse()` output is preferable to `str()` because it provides a concise overview of variable types while remaining easy to read.

Step 2.3 — Select the Required Variables

The assignment specifies exactly 15 variables.

```
ndhs <- ndhs %>%
  select(
    hw71,
    hw72,
    hw1,
    b4,
    b5,
```

```

    m4,
    v024,
    v025,
    v106,
    v701,
    v130,
    v190,
    v136,
    v137,
    v005
  )

```

Selecting only the required variables:

- reduces memory usage,
- simplifies debugging,
- improves reproducibility.

Step 2.4 — Create Clean WAZ and WHZ Variables

This is the most important cleaning step.

```

ndhs <- ndhs %>%
  mutate(

    waz = case_when(
      hw71 >= 9996 ~ NA_real_,
      TRUE ~ hw71 / 100
    ),

    whz = case_when(
      hw72 >= 9996 ~ NA_real_,
      TRUE ~ hw72 / 100
    )
  )

```

Notice that we **do not overwrite** the original DHS variables.

Instead, we preserve:

```

hw71
hw72

```

and create

```
waz  
whz
```

This is considered good research practice because the raw variables remain available for auditing.

Step 2.5 — Recode Child Sex

```
ndhs <- ndhs %>%  
  mutate(  
  
    sex = recode(  
      as.character(b4),  
  
      "1" = "Male",  
      "2" = "Female"  
    )  
  
  )
```

Step 2.6 — Child Survival

```
ndhs <- ndhs %>%  
  mutate(  
  
    child_alive = recode(  
      as.character(b5),  
  
      "0" = "No",  
      "1" = "Yes"  
    )  
  
  )
```

Step 2.7 — Breastfeeding

The assignment explicitly instructs us to collapse breastfeeding into **three groups**.

```
ndhs <- ndhs %>%  
  mutate(  
  
    breastfeeding = case_when(  
  
      m4 == 93 ~ "Never",  
  
      m4 == 94 ~ "Still Breastfeeding",  
  
      m4 == 95 ~ "Stopped",  

```

```

TRUE ~ NA_character_

)

)

```

This follows the assignment specification exactly.

Step 2.8 — Region

```

ndhs <- ndhs %>%
  mutate(

    region = recode(

      as.character(v024),

      "1" = "North Central",

      "2" = "North East",

      "3" = "North West",

      "4" = "South East",

      "5" = "South South",

      "6" = "South West"

    )

  )

```

Step 2.9 — Residence

```

ndhs <- ndhs %>%
  mutate(

    residence = recode(

      as.character(v025),

      "1" = "Urban",

      "2" = "Rural"

    )

  )

```

```
)
```

Step 2.10 — Mother's Education

```
ndhs <- ndhs %>%  
  mutate(  
  
    mother_education = recode(  
  
      as.character(v106),  
  
      "0" = "None",  
  
      "1" = "Primary",  
  
      "2" = "Secondary",  
  
      "3" = "Higher"  
  
    )  
  
  )
```

Step 2.11 — Father's Education

The assignment specifies that code **8 ("Don't know")** should be converted to NA.

```
ndhs <- ndhs %>%  
  mutate(  
  
    father_education = case_when(  
  
      v701 == 8 ~ NA_character_,  
  
      v701 == 0 ~ "None",  
  
      v701 == 1 ~ "Primary",  
  
      v701 == 2 ~ "Secondary",  
  
      v701 == 3 ~ "Higher"  
  
    )  
  
  )
```

Step 2.12 — Religion

The assignment instructs us to collapse categories into **Christianity**, **Islam**, and **Other**.

```
ndhs <- ndhs %>%
  mutate(

    religion = case_when(

      v130 == 1 ~ "Christianity",

      v130 == 2 ~ "Islam",

      TRUE ~ "Other"

    )

  )
```

Important: Before finalizing this section, we should inspect the actual coding of v130 in your dataset with `table(v130, useNA = "ifany")`. DHS coding can vary slightly by country and survey year, so it's better to verify than assume.

Step 2.13 — Ordered Factors

This is one area where many students lose marks.

These variables are **ordinal**, not nominal.

```
ndhs <- ndhs %>%
  mutate(

    wealth = factor(

      v190,

      levels = 1:5,

      labels = c(

        "Poorest",

        "Poorer",

        "Middle",

        "Richer",
```

```

    "Richest"

  ),

  ordered = TRUE

),

mother_education = factor(

  mother_education,

  levels = c(

    "None",

    "Primary",

    "Secondary",

    "Higher"

  ),

  ordered = TRUE

)

)

```

Ordered factors preserve the natural ranking, which is especially useful for summaries and non-parametric analyses.

Step 2.14 — Data Quality Summary

The assignment asks for:

- total rows,
- valid WAZ observations,
- valid WHZ observations,
- percentage missing.

A clean solution is:

```
cat("Total observations:", nrow(ndhs), "\n")
```

```
cat("Valid WAZ:", sum(!is.na(ndhs$waz)), "\n")
```

```
cat("Valid WHZ:", sum(!is.na(ndhs$whz)), "\n")
```

```
cat(
  "WAZ Missing:",
  round(mean(is.na(ndhs$waz))*100,2),
  "%\n"
)
```

```
cat(
  "WHZ Missing:",
  round(mean(is.na(ndhs$whz))*100,2),
  "%\n"
)
```

Research-Level Improvement

Instead of stopping there, I recommend generating a **Data Quality Dashboard** that summarizes the entire cleaned dataset.

```
quality_summary <- tibble(
  Variable = names(ndhs),
  Missing = colSums(is.na(ndhs)),
  Missing_Percent = round(colMeans(is.na(ndhs)) * 100, 2),
  Data_Type = sapply(ndhs, class)
) %>%
  arrange(desc(Missing_Percent))
```

```
quality_summary %>%
  kable(caption = "Data Quality Summary") %>%
  kable_styling(full_width = FALSE)
```

This goes beyond the assignment requirements but demonstrates research maturity and provides a reusable diagnostic table for future analyses.

Milestone Check

At the end of Task A, you should have:

- ✓ Dataset loaded.
- ✓ Fifteen required variables selected.
- ✓ Clean waz and whz variables created.
- ✓ DHS missing codes converted to NA.
- ✓ Readable categorical labels.

- ✓ Ordered factors for wealth and education.
- ✓ Data quality summary generated.

From here, we will proceed to **Task B (Descriptive Statistics)**, where we'll produce publication-quality tables with interpretations rather than simple numeric outputs. We will also verify a few variable codings (especially v130) against your actual dataset before locking in the final script to ensure complete accuracy.

We now enter what I consider the **heart of EDA**. Any software can compute descriptive statistics, but a data scientist must be able to **extract meaningful insights** from them.

Our objective is not just to satisfy **Task B**, but to produce publication-quality tables and interpretations.

Phase 3 — Task B

Descriptive Statistics

The assignment requires five tables. We'll produce them in a logical order.

Table	Purpose
Table 1	Data completeness (missing values)
Table 2	Summary statistics
Table 3	National prevalence
Table 4	Comparison of HAZ, WAZ and WHZ
Table 5	Underweight prevalence across subgroups

This ordering tells a coherent story:

Can we trust the data → What does the data look like → How common is malnutrition → Which type is most common → Who is most affected?

Table 1

Missing Value Analysis

This is one table that many published papers omit, but reviewers appreciate because it demonstrates transparency.

R Code

```
#####
# Missing Value Table
#####

missing_table <- ndhs %>%
  summarise(across(everything(),
    ~sum(is.na(.)))) %>%
  pivot_longer(
```

```

    everything(),
    names_to="Variable",
    values_to="Missing"
  ) %>%
  mutate(

    Percent = round(
      Missing/nrow(ndhs)*100,
      2
    )

  ) %>%
  arrange(desc(Percent))

```

missing_table

Better presentation

```
missing_table %>%
```

```
kable(
```

```
caption="Table 1. Missing Data Summary"
```

```
) %>%
```

```
kable_styling(
  full_width=FALSE
)
```

Interpretation

Instead of writing

WAZ had 4.1% missing values.

Write something like:

Missingness was generally low across the variables selected for this analysis. Anthropometric indicators exhibited only a small proportion of missing observations after DHS invalid codes were recoded to missing values. This suggests that the dataset is sufficiently complete for reliable exploratory analyses, although any variables with comparatively higher missingness should be interpreted cautiously.

Notice:

No numbers are invented.

We'll insert the actual percentages after running the analysis.

Table 2

Summary Statistics

The assignment requests

- Mean
- Median
- Standard deviation
- Minimum
- Maximum
- Skewness

for

- WAZ
- WHZ

R Code

```
#####  
# Summary Statistics  
#####  
  
summary_table <-  
  
tibble(  
  
  Variable=c("WAZ","WHZ"),  
  
  Mean=c(  
    mean(ndhs$waz,na.rm=TRUE),  
    mean(ndhs$whz,na.rm=TRUE)  
  ),  
  
  Median=c(  
  
    median(ndhs$waz,na.rm=TRUE),  
  
    median(ndhs$whz,na.rm=TRUE)  
  
  ),  
  
  SD=c(  

```

```

sd(ndhs$waz,na.rm=TRUE),

sd(ndhs$whz,na.rm=TRUE)

),

Minimum=c(

min(ndhs$waz,na.rm=TRUE),

min(ndhs$whz,na.rm=TRUE)

),

Maximum=c(

max(ndhs$waz,na.rm=TRUE),

max(ndhs$whz,na.rm=TRUE)

),

Skewness=c(

e1071::skewness(ndhs$waz,
na.rm=TRUE),

e1071::skewness(ndhs$whz,
na.rm=TRUE)

)

)

summary_table

Better formatting

summary_table %>%

mutate(

across(where(is.numeric),
round,
2)

```

```
) %>%
```

```
kable(  
caption="Summary Statistics"  
)
```

Interpretation

This is where many students lose marks.

A weak interpretation:

The mean WAZ is -0.81.

A stronger interpretation:

Both anthropometric indicators were centred below the WHO reference mean of zero, suggesting that Nigerian children in the sample generally experienced poorer nutritional status than the international reference population. Differences between the mean and median, together with the skewness coefficients, provide evidence regarding the symmetry of each distribution and indicate whether robust statistics such as the median may be more appropriate than the mean.

Notice that we are **interpreting** rather than merely repeating numbers.

Table 3

National Prevalence

Create binary variables.

```
ndhs <-
```

```
ndhs %>%
```

```
mutate(  
  
underweight=  
waz<-2,  
  
severe_underweight=  
waz<-3,  
  
wasting=  
whz<-2,  
  
severe_wasting=  
whz<-3
```

)

Now calculate prevalence.

```
prevalence_table<-
```

```
tibble(
```

```
Indicator=c(
```

```
"Underweight",
```

```
"Severe Underweight",
```

```
"Wasting",
```

```
"Severe Wasting"
```

```
),
```

```
Percent=c(
```

```
mean(ndhs$underweight,  
na.rm=TRUE),
```

```
mean(ndhs$severe_underweight,  
na.rm=TRUE),
```

```
mean(ndhs$wasting,  
na.rm=TRUE),
```

```
mean(ndhs$severe_wasting,  
na.rm=TRUE)
```

```
)*100
```

```
)
```

Interpretation

Here we begin answering

How serious is malnutrition nationally?

Example structure

Approximately XX% of Nigerian children were classified as underweight, whereas YY% were classified as severely underweight. Wasting affected ZZ% of children, with severe wasting occurring less frequently. These findings indicate that both chronic and acute forms of malnutrition remain important public health concerns.

Table 4

Comparing HAZ, WAZ and WHZ

Although the assignment focuses on WAZ and WHZ, it explicitly asks for comparison with HAZ from the class exercise.

Instead of manually copying results, we'll compute all three in a consistent way (assuming HAZ has been prepared as haz in your earlier class work).

```
comparison_table<-
```

```
tibble(
```

```
Condition=c(
```

```
"Stunting",
```

```
"Underweight",
```

```
"Wasting"
```

```
),
```

```
Percent=c(
```

```
mean(ndhs$haz<-2,  
na.rm=TRUE),
```

```
mean(ndhs$underweight,  
na.rm=TRUE),
```

```
mean(ndhs$wasting,  
na.rm=TRUE)
```

```
)*100
```

```
)
```

Better presentation

```
comparison_table %>%
```

```

arrange(
  desc(Percent)
)

```

Immediately, you'll know which nutritional condition is most prevalent.

Table 5

Underweight by Important Groups

This is the first real epidemiological analysis.

We'll compute prevalence by:

- Region
- Residence
- Wealth
- Mother's education
- Religion

We'll create a reusable helper function instead of repeating code.

```

#####
# Reusable prevalence function
#####

calculate_prevalence <- function(data, group_var){

  data %>%

  group_by({{group_var}}) %>%

  summarise(

    Children=n(),

    Underweight=

    mean(
      underweight,
      na.rm=TRUE
    )*100,

    .groups="drop"

  ) %>%

```

```
arrange(  
  desc(Underweight)  
)
```

```
}
```

Now we simply call

```
table_region<-
```

```
calculate_prevalence(  
  ndhs,  
  region  
)
```

```
table_wealth<-
```

```
calculate_prevalence(  
  ndhs,  
  wealth  
)
```

```
table_education<-
```

```
calculate_prevalence(  
  ndhs,  
  mother_education  
)
```

```
table_religion<-
```

```
calculate_prevalence(  
  ndhs,  
  religion  
)
```

```
table_residence<-
```

```
calculate_prevalence(  
  ndhs,  
  residence  
)
```

This approach is modular, easier to maintain, and avoids duplicating nearly identical code blocks.

Research Tip

Instead of merely identifying the subgroup with the highest prevalence, ask:

- Is there a socioeconomic gradient?
- Does underweight consistently decrease with increasing wealth?
- Is there an educational gradient?
- Are regional differences substantial enough to motivate further statistical testing?

These questions naturally lead into the research questions required in Task E and demonstrate critical thinking beyond descriptive reporting.

Software Engineering Improvement

The instructor's sample script is intentionally linear for teaching purposes. Since one of our goals is to build a reusable research portfolio, I recommend we refactor the project slightly by creating a `functions.R` file containing helper functions such as:

- `calculate_prevalence()`
- `summary_statistics()`
- `missing_value_summary()`

Your main `Sprint01_BasilEmeokoro.R` script then becomes a clear sequence of analysis steps rather than a long monolithic script. This improves readability, reusability, and maintainability while still satisfying the assignment requirements. It also aligns with the software engineering standards you established at the start of this project.